

A Study on Pathological Myopia Diagnosis Practices in Kampala

By

BSE 25-4

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Declaration

We declare that this report, titled "A Study on Pathological Myopia Diagnosis Practices in Kampala," is the result of our completed research and collaborative efforts. Each group member has contributed equally to the research, analysis, and preparation of this document.

We confirm that the work is original, has not been submitted elsewhere, and adhere to academic integrity standards. All sources and references have been properly acknowledged.

This report reflects our commitment to understanding and addressing the challenges of diagnosing pathological myopia in Kampala, and we hope our findings contribute valuable insights to the medical field.

Acknowledgement

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Abstract

Pathological myopia, a severe form of nearsightedness characterized by excessive axial elongation of the eye, affects approximately 3% of the global population and poses significant challenges in clinical settings due to its progressive and often irreversible nature. This condition, which frequently manifests between the ages of 6 and 13, continues to progress throughout life, leading to potential vision loss during individuals' most productive years. In Uganda, particularly in Kampala, diagnostic practices for pathological myopia rely heavily on conventional manual methods, which often result in inaccurate diagnoses and delayed interventions.

This study evaluates the current diagnostic landscape in Kampala, identifies gaps in practice, and explores the potential for integrating advanced technologies such as artificial intelligence (AI) to improve diagnostic accuracy. Utilizing a mixed-methods approach, the research involved a comprehensive review of existing literature, analysis of clinical data, and interviews with ophthalmology professionals. Findings indicate a notable absence of standardized diagnostic protocols, limited access to advanced tools, and significant barriers to adopting AI-based solutions, including infrastructural limitations and insufficient training for healthcare providers.

The results of this study underscore the urgent need for enhanced diagnostic practices and the adoption of innovative technologies in managing pathological myopia in Uganda. By integrating AI-driven solutions, healthcare systems can achieve earlier detection, improved diagnostic precision, and better patient outcomes. This report provides a foundational step toward modernizing the diagnostic approach to pathological myopia, ultimately contributing to the mitigation of vision loss in affected populations.

1. Introduction

1.1 Background

Pathological myopia is a significant public health concern, particularly in developing countries like Uganda, where it contributes to a broader spectrum of visual impairments. Defined by excessive axial elongation of the eye, pathological myopia affects approximately 3% of the global population and can lead to severe vision loss that is often irreversible. This condition typically manifests during childhood, progressing throughout an individual's life and impacting their ability to function during crucial productive years.

In Uganda, the situation is exacerbated by a critical shortage of trained eye care professionals. As of 2017, the country had only 48 ophthalmologists to serve a population of over 42 million, with many of these specialists concentrated in the capital city, Kampala. This disparity means that a large portion of the population lacks access to specialized eye care services, relying instead on non-ophthalmologists who may not have the expertise necessary to diagnose and manage complex conditions like pathological myopia effectively [1][3].

According to the data we collected from health practitioners, all have experience of less than 5 years in ophthalmology, with only 22% having more than 3 years of experience. This causes a significant challenge given the fact that accuracy in manual diagnosis highly depends on the experience one has in diagnosing the disease.

The reliance on conventional manual diagnostic methods in Kampala has resulted in significant challenges. Many eye conditions remain undiagnosed or inadequately treated due to outdated practices and a lack of standardized protocols. Consequently, patients often experience delays in receiving appropriate care, leading to preventable vision loss. The Ministry of Health has acknowledged that uncorrected refractive errors are among the leading causes of visual impairment in Uganda, yet there remains an urgent need for more effective diagnostic tools and training for healthcare providers [1][4].

Whereas 66.7% and 33.3% of the health practitioners consider early diagnosis as a very effective and moderately effective way in preventing severe complications of pathological myopia respectively, over 47.2% of the sample in the general public we collected never go for medical checkups and 40.2% only do so when they have an issue.

What is missing from the current landscape is the integration of advanced technologies, particularly artificial intelligence (AI), into the diagnostic process for pathological myopia. The absence of standardized diagnostic protocols and limited access to modern diagnostic tools hinder timely and accurate diagnoses.

Furthermore, infrastructural limitations and insufficient training for healthcare providers pose substantial barriers to adopting AI-driven solutions.

To address these challenges, this report proposes the integration of AI technologies into the diagnostic framework for pathological myopia in Uganda. By leveraging AI's capabilities for enhanced diagnostic accuracy and efficiency, healthcare systems can facilitate earlier detection and better management of this progressive condition. This approach aims not only to improve patient outcomes but also to modernize the overall eye care landscape in Uganda, ultimately reducing the burden of vision loss associated with pathological myopia.

1.2 Problem statement

Pathological myopia, a severe form of nearsightedness characterized by excessive axial elongation of the eye, poses significant challenges to vision health, affecting approximately 3% of the global population. In Uganda, particularly in urban areas like Kampala, the diagnostic practices for this condition primarily rely on conventional manual methods. These traditional approaches often lead to inaccurate diagnoses and delayed interventions, resulting in preventable vision loss among individuals during their most productive years.

While some efforts have been made to address the challenges associated with diagnosing pathological myopia such as training healthcare professionals and increasing awareness of the condition—these initiatives have not been sufficient to establish standardized diagnostic protocols or improve access to advanced diagnostic technologies.

What is missing, however, is a comprehensive integration of innovative solutions, particularly artificial intelligence (AI), into the diagnostic framework for pathological myopia in Uganda. The lack of advanced tools and standardized practices limits the ability of healthcare providers to accurately diagnose and manage this progressive condition effectively. This not only compromises the accuracy of diagnoses but also hinders timely treatment interventions. As a result, individuals may experience significant vision impairment that could have been mitigated with earlier detection and appropriate management.

To address these gaps, this study proposes the integration of AI-driven diagnostic solutions into ophthalmology practices in Kampala. By leveraging advanced technologies, healthcare systems can enhance diagnostic accuracy, facilitate earlier detection of pathological myopia, and ultimately improve patient outcomes. This approach aims to modernize the diagnostic landscape and mitigate the impact of vision loss on affected populations.

1.3 Objectives

1.3.1 General Objective

To evaluate the current diagnostic practices for pathological myopia in Kampala and explore the potential integration of advanced technologies, such as artificial intelligence (AI), to enhance diagnostic accuracy.

To achieve this overarching goal, the report delineates specific objectives:

1.3.2 Specific objectives

To assess the existing diagnostic protocols for pathological myopia in clinical and hospital settings across Kampala.

To identify barriers to the adoption of advanced diagnostic tools and AI technologies within ophthalmology practices.

To explore healthcare professionals' perspectives on integrating AI-driven solutions into current diagnostic frameworks.

To propose recommendations for improving diagnostic accuracy and patient outcomes through innovative technological interventions.

2. Literature Review

The integration of artificial intelligence (AI) in the diagnosis of pathological myopia has gained substantial attention, reflecting the potential of machine learning and deep learning techniques to revolutionize diagnostic practices. Pathological myopia, characterized by excessive axial elongation of the eye and associated structural changes, remains a complex condition, particularly given the subjective variability of traditional diagnostic methods. Studies highlight the transformative role of AI in enhancing diagnostic precision, efficiency, and accessibility.

2.1 AI Applications and Advancements

Multiple studies underscore the efficacy of AI in diagnosing pathological myopia. Zhang et al. (2013) introduced a computer-aided diagnosis framework employing multiple kernel learning (MKL) to integrate heterogeneous biomedical data, achieving significant improvements in diagnostic accuracy. Similarly, Kim et al. (2021) demonstrated machine learning algorithms' effectiveness in predicting pathological myopia by analyzing tomographic elevation of the posterior sclera. Recent meta-analyses by Zhang et al. (2023) further emphasize the diagnostic accuracy of AI, reporting pooled sensitivity and specificity rates of 95% and 97%, respectively, for detecting pathological myopia using fundus and optical coherence tomography (OCT) images.

Deep learning has been pivotal in this domain, particularly convolutional neural networks (CNNs) such as ResNet and DenseNet. Hemelings et al. (2021) successfully applied deep learning for lesion segmentation and classification using fundus images, while Devda and Eswari (2019) achieved high performance in image analysis tasks, achieving AUC scores of 0.95 or higher. Jiang et al. (2019) introduced conditional generative adversarial networks (cGANs) for lesion segmentation, achieving 96% accuracy through innovative network designs that address data imbalance issues.

2.2 Datasets and Methodologies

The establishment of annotated datasets like the PALM dataset, consisting of 1,200 fundus images, has enabled benchmarking and reproducibility in AI-based pathological myopia research. Transfer learning techniques have further optimized performance in scenarios with limited training data, as demonstrated by Xie et al. (2019) and Chen et al. (2021). In addition, randomized algorithms, such as Lin et al.'s (2020) application of Random Forest with cross-validation methods, achieved accuracies of up to 85%, showcasing the diverse approaches in AI-driven diagnostics.

2.3 Traditional vs. AI-Based Diagnostics

Traditional diagnostic methods for pathological myopia, involving manual fundus image analysis, are time-intensive, requiring up to 4–5 hours for completion. In contrast, AI and deep learning technologies have reduced this time to seconds, significantly enhancing clinical efficiency. AI approaches, such as ResNet-50 and VGG19 models, have not only automated lesion detection but also facilitated tasks like the identification of retinal vessel patterns and fovea localization, addressing key gaps in conventional methodologies.

2.4 Challenges and Opportunities

While the potential of AI in pathological myopia diagnosis is evident, several challenges remain, particularly in resource-limited settings like Uganda. The exploration of AI for myopia diagnosis in Uganda is limited, with broader studies on AI adoption in Africa highlighting infrastructural deficits, governance issues, and a lack of local expertise as critical barriers (Nibigira, 2021; Babarinde, 2020). Nonetheless, the literature suggests that addressing these challenges could unlock AI's transformative potential for ophthalmology in the region.

Globally, the need for external validation datasets and the integration of AI systems into clinical workflows remains a priority to ensure their reliability and scalability (Tang et al., 2022). Discussions by Zhang and Zou (2021) on AI's relevance in addressing global myopia highlight the importance of standardized methodologies and infrastructure improvements to realize AI's full potential in diagnostics.

3. Methodology

3.1 Research Philosophy

This study of Pathological myopia practices in Kampala followed a pragmatic research philosophy, which combines both quantitative and qualitative methods. This approach was chosen to provide a complete understanding of the topic by integrating insights from both the general public and eye care professionals(ophthalmologists).

3.2 Research Approach

A mixed-methods approach was used, combining quantitative data from questionnaires and qualitative data from interviews. This combination allowed us to analyze statistical trends while also understanding the deeper context behind the data.

3.3 Research Strategy

The research employed two main strategies: **Surveys:** Online questionnaires were used to collect data from a large number of respondents. **Interviews:** In-person interviews were conducted to explore specific themes in greater depth.

3.4 Data Collection Methods

Online Questionnaires: Two separate questionnaires were created using Google Forms: one for the general public and one for eye care professionals. The general public questionnaire received 127 responses, while the questionnaire for medical professionals had 9 responses. Questions included a mix of closed-ended (quantitative) and open-ended (qualitative) formats.

Interviews: Two face-to-face interviews were conducted with eye care professionals at Makerere University Hospital and Mulago Hospital. These interviews explored topics that emerged from the questionnaire responses and provided deeper insights.

3.5 Data Analysis

Quantitative Data: Statistical analysis was conducted on questionnaire responses to identify trends and patterns. Frequencies, percentages, and comparisons between the general public and professionals were calculated.

Qualitative Data: Responses to open-ended questions and interview transcripts were analyzed thematically. Key themes were identified to enrich the understanding of the issues.

3.6 Sampling

General Public: Convenience sampling was used, sharing the questionnaire via social media, email, and personal networks to reach participants easily.

Eye Care Professionals: Purposive sampling targeted medical professionals working in the field to ensure relevant expertise and insights.

3.7 Data Collection Locations

Questionnaires were distributed online through platforms like social media and email. The interviews were conducted in person at Makerere University Hospital and Mulago Hospital, providing a relevant setting for discussions with medical professionals.

4. Results

4.1 Results for the general public

4.1.1 Demographics

Age Bracket Distribution

Age Bracket	Responses	Percentage (%)
18-24 years	96	75.6%
25-34 years	27	21.3%
35-44 years	3	2.4%
55 years & above	1	0.8%

Table 1 Age Bracket Distribution

Age Distribution table

Visualization: Figure 1 shows Age Distribution

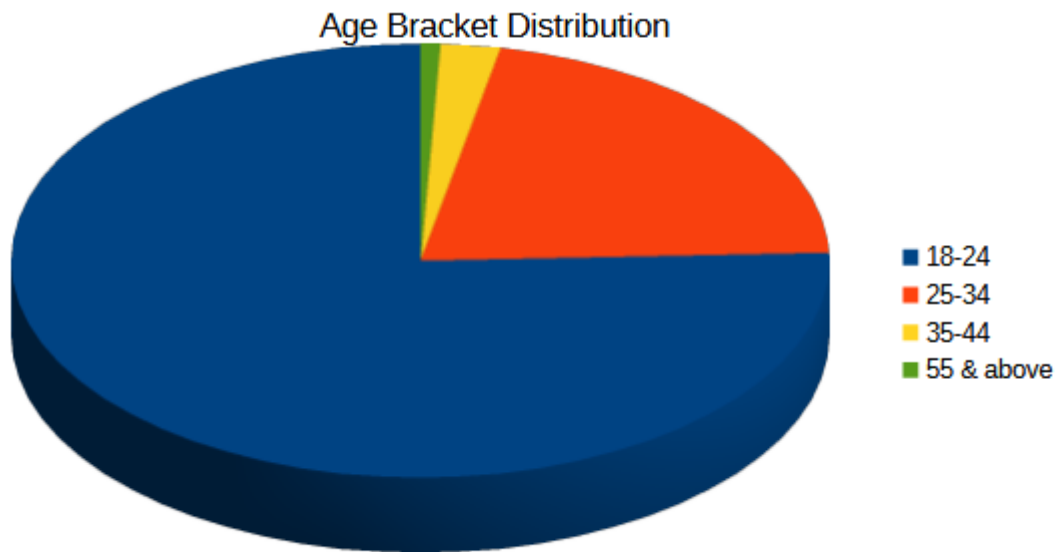


Figure 1 Age Distribution

4.1.2 Glasses/Contact Lens Usage

Response	Count	Percentage (%)
No	102	80.3
Yes	25	19.7

Table 2 Glasses/Contact Lens Usage

Narrative: 80.3% of respondents indicated they do not wear glasses or contact lenses, while 19.7% reported using them.

Visualization: Figure 2 shows Contact Lense Usage.

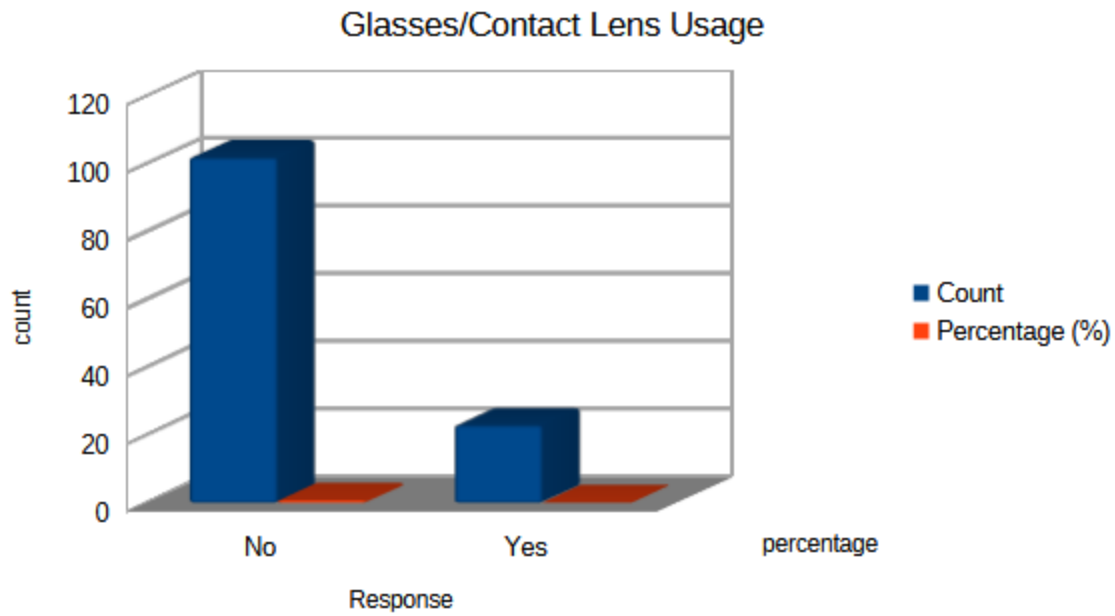


Figure 2 Contact Lense Usage.

4.1.3 Diagnosis of Myopia

Response	Count	Percentage (%)
No	108	85
Yes	19	15

Table 3 Diagnosis of Myopia

Narrative: Most respondents (85%) have not been diagnosed with myopia, whereas 15% indicated they have been diagnosed.

Visualization: Figure 3 shows the rate of myopia diagnosis.

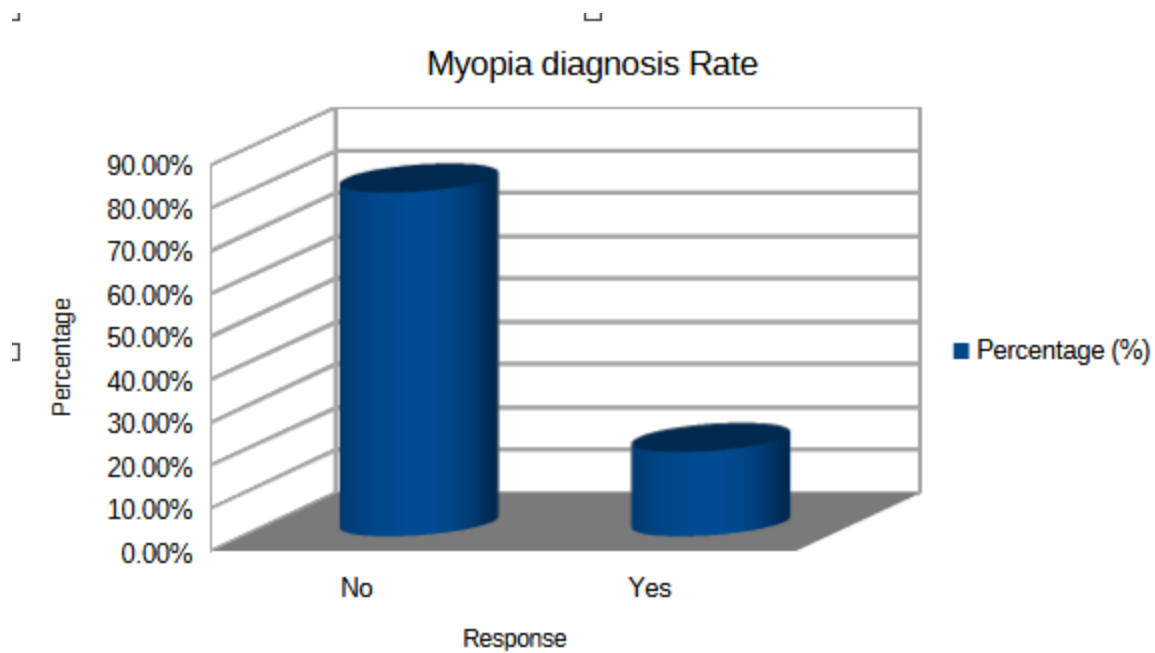


Figure 3 The rate of myopia diagnosis.

4.1.4 Daily Screen Time

Response	Count	Percentage (%)
Yes	117	92.1
No	10	7.9

Table 4 Daily Screen Time

Narrative: A significant portion of respondents (92.1%) reported spending more than 4 hours a day looking at screens, highlighting widespread screen exposure.

Visualization: Figure 4 shows Screen time of the public.

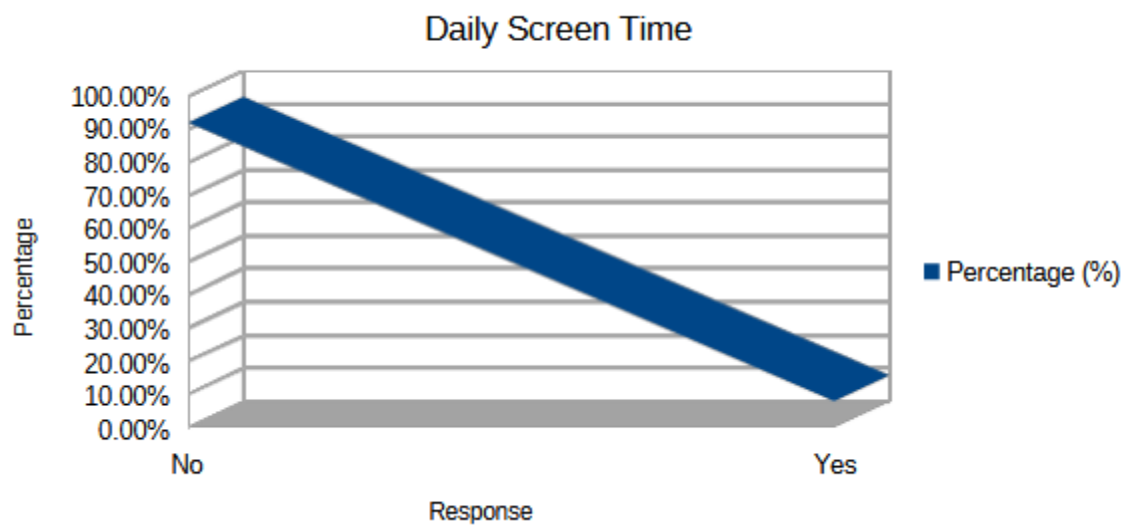


Figure 4 Screen time of the public.

4.1.5 Frequency of Eye Checkups

Frequency	Count	Percentage (%)
Once a year	11	8.7
Every 2-3 years	5	3.9
Only when I have an issue	51	40.2
Never	60	47.2

Table 5 Frequency of Eye Checkups

Visualization: figure 5 shows Frequency of Eye Checkups.

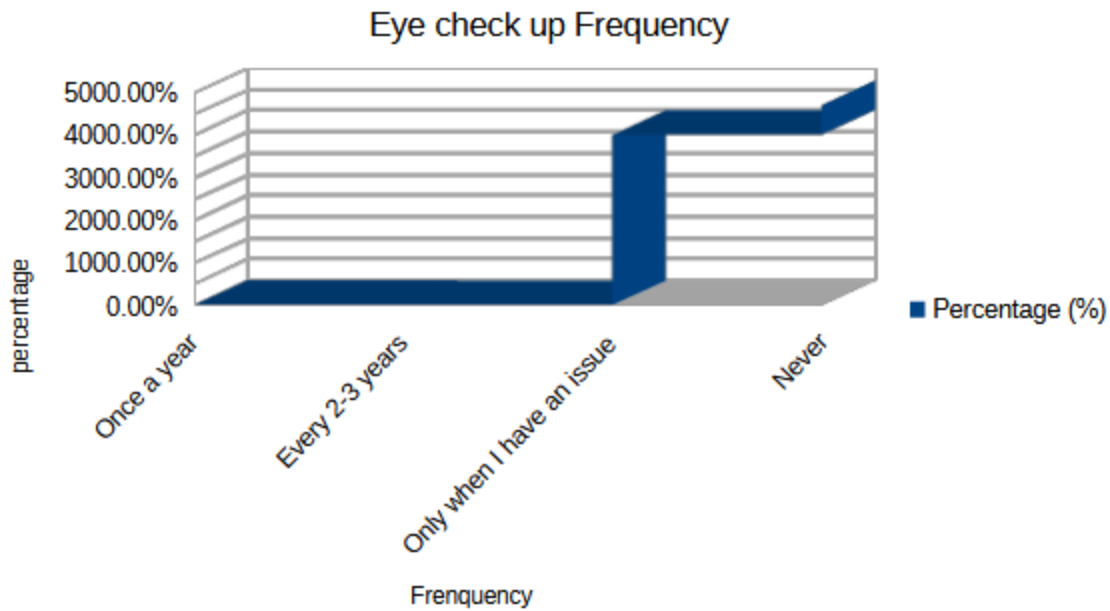


Figure 5 Frequency of Eye Checkups

Narrative: Nearly half of respondents (47.2%) never go for eye checkups, and 40.2% visit an eye doctor only when they experience an issue. A small fraction (8.7%) visits annually, while 3.9% check every 2-3 years.

4.1.6 Awareness of Pathological Myopia

Response	Count	Percentage (%)
Yes	62	48.8
No	65	51.2

Table 6 Awareness of Pathological Myopia

Visualization: figure 6 illustrates awareness of pathological myopia among the general public.

Awareness of Pathological Myopia

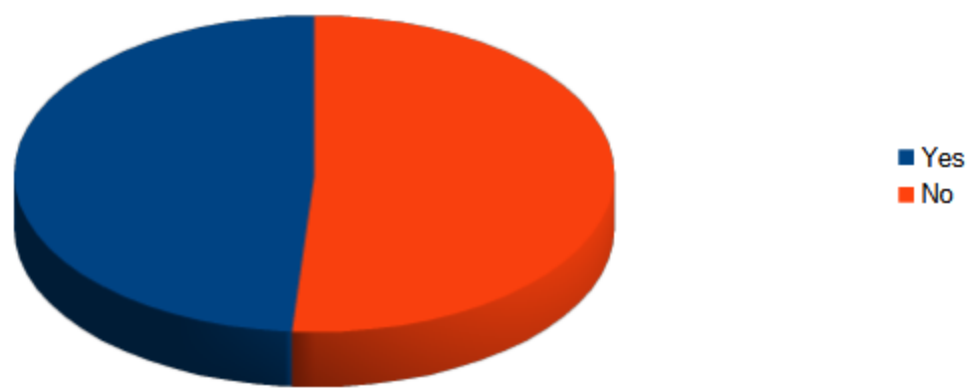


Figure 6 Awareness of Pathological Myopia

Narrative: Approximately half of the respondents (51.2%) had not heard of pathological myopia, while 48.8% were aware of it.

Sources of Awareness

Source	Count	Percentage (%)
Internet	27	35.1
Doctor	15	19.5
Family/Friends	10	13
School	3	3.9

Others (misc.)	7	9.1
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Table 7 Sources of Awareness

Narrative: The most common source of awareness was the Internet (35.1%), followed by doctors (19.5%) and family/friends (13%).

Visualization: figure 7 illustrates awareness.



Figure 7 Sources of Awareness

4.1.7 Symptoms Experienced

Symptom	Count	Percentage (%)
Eye strain	55	43.3
Blurred vision	36	28.3
Difficulty seeing distant	32	25.2

objects		
Floaters	9	7.1
None of the above	53	41.7

Table 8 Symptoms Experienced

Narrative: Eye strain (43.3%) and blurred vision (28.3%) were the most reported symptoms. A notable 41.7% of respondents reported experiencing no symptoms.

4.1.8 Eye Care Practices

Practice	Count	Percentage (%)
Take breaks from screens	75	59.1
Eat a diet rich in eye-healthy nutrients	50	39.4
Wear glasses/contact lenses as prescribed	21	16.5
Visit an eye doctor periodically	17	13.4
None of the above	22	17.3

Table 9 Eye Care Practices

Visualization: figure 8 shows eye care practices.

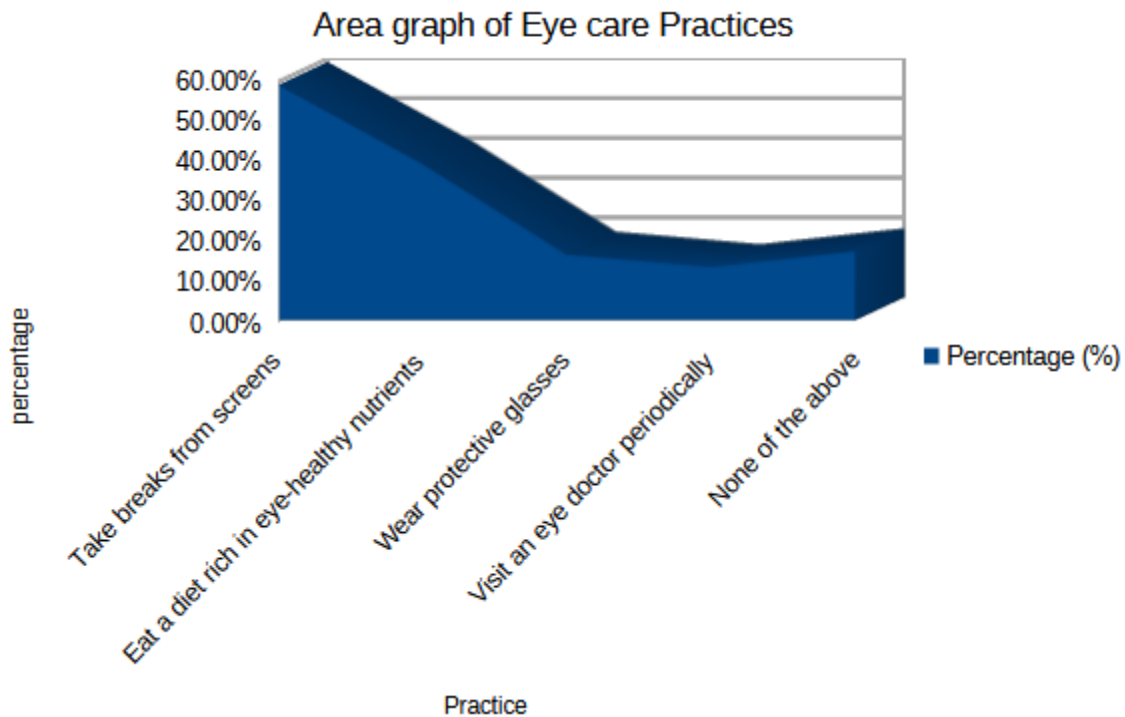


Figure 8 Eye care practices

Narrative: Taking breaks from screens (59.1%) was the most common practice for maintaining eye health, followed by a nutrient-rich diet (39.4%).

4.2 Results for Eye Care Specialists

4.2.1 Professional Experience and Encounters with Pathological Myopia

The majority of respondents (66.7%) had 1–3 years of experience in ophthalmology, while 22.2% had less than one year. Only 11.1% reported having 3–5 years of experience. Most (88.9%) had encountered patients with Pathological Myopia in their practice, while 11.1% had not

Experience Level	Frequency (%)	Encountered Pathological Myopia	Frequency (%)
Less than 1 year	22.2	Yes	88.9
1–3 years	66.7	No	11.1

3–5 years	11.1		
5–10 years	0		
More than 10 years	0		

Table 10 Professional Experience and Encounters with Pathological Myopia

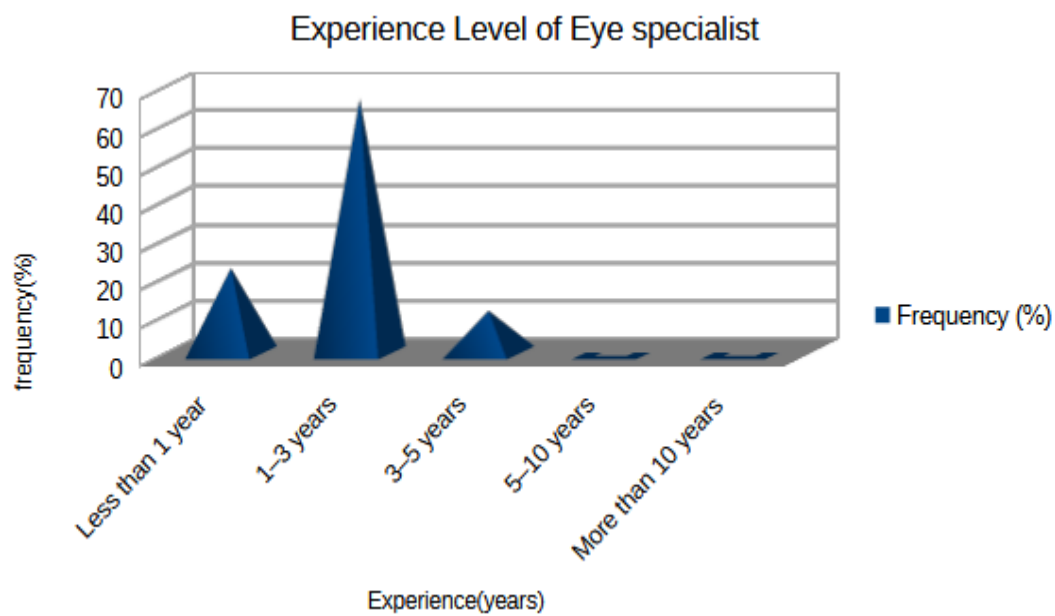


Figure 9 Experience of Eye Specialists

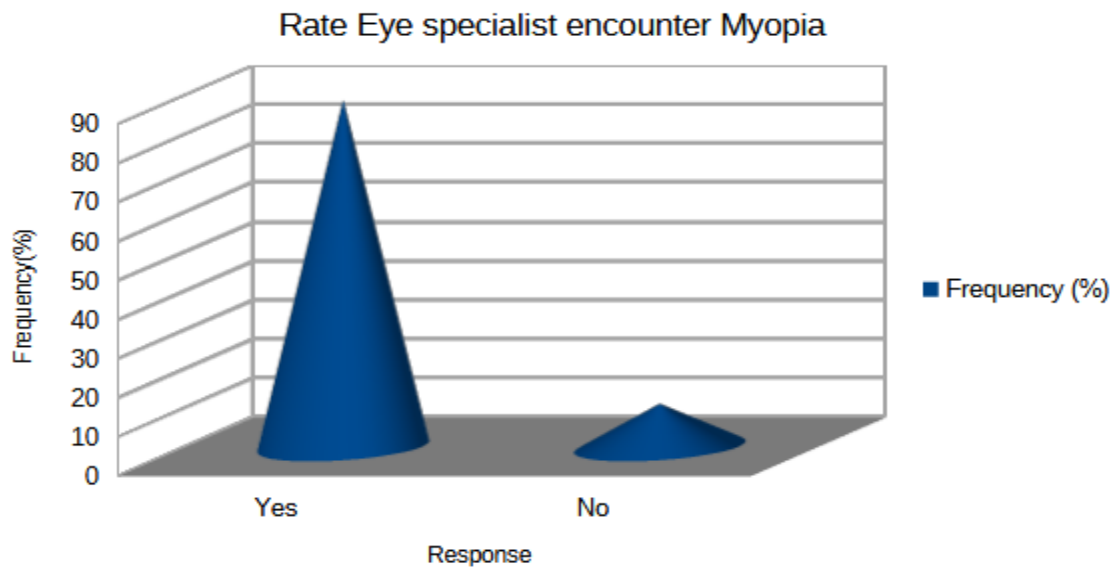


Figure 10 Rate Eye Specialist Encounter

4.2.2 Frequency of Diagnosing Pathological Myopia

The frequency of diagnosing Pathological Myopia was split between those who diagnose it *rarely* (55.6%) and *frequently* (55.6%), with one respondent (11.1%) diagnosing it *occasionally*.

Frequency of Diagnosis	Frequency (%)
Rarely	55.6
Occasionally	11.1
Frequently	33.3

Table 11 Frequency of Diagnosing Pathological Myopia

figure 11 represents the frequency of Diagnosing Pathological Myopia.

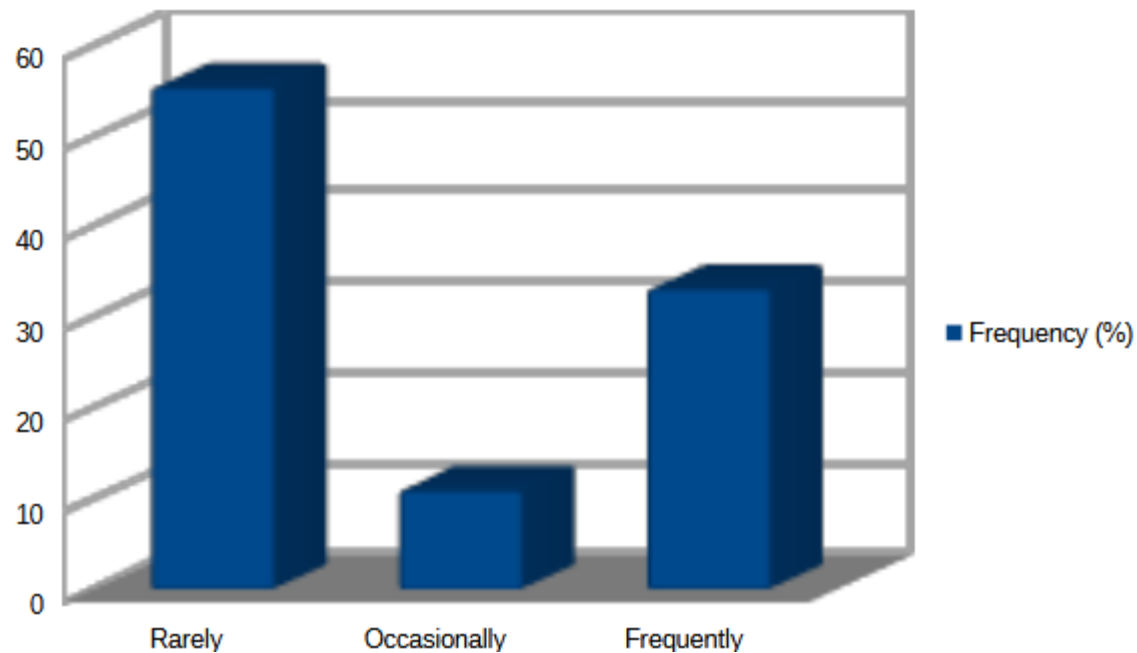


Figure 11 Frequency of Diagnosing PM

4.2.3 Common Symptoms and Risk Factors

All respondents (100%) associated blurred vision with Pathological Myopia. Other symptoms included retinal detachment (66.7%), eye strain (55.6%), and floaters (11.1%). The top risk factors were prolonged near work (77.8%), genetics (66.7%), and lack of outdoor activities (55.6%)

Symptoms	Frequency (%)	Risk Factors	Frequency (%)
Blurred vision	100	Prolonged near work	77.8
Retinal detachment	66.7	Genetics	66.7
Eye strain	55.6	Lack of outdoor activities	55.6
Floaters	11.1	Eye diseases or trauma	44.4

Table 12 Common Symptoms and Risk Factors

4.2.4 Diagnostic Tools and Effectiveness of Early Diagnosis

The most commonly used diagnostic tool was fundus photography (44.4%), followed by optical coherence tomography (OCT) (33.3%) and retinal imaging (22.2%). Most respondents (66.7%) considered early diagnosis *very effective*, while 33.3% rated it as *moderately effective*.

Diagnostic Tools	Frequency (%)	Effectiveness of Early Diagnosis
Fundus photography	44.4	Very effective
Optical Coherence Tomography	33.3	Moderately effective
Retinal imaging	22.2	Effective

Table 13 Diagnostic Tools and Effectiveness of Early Diagnosis

figure 12 represents the tools used in Diagnosing Pathological Myopia.

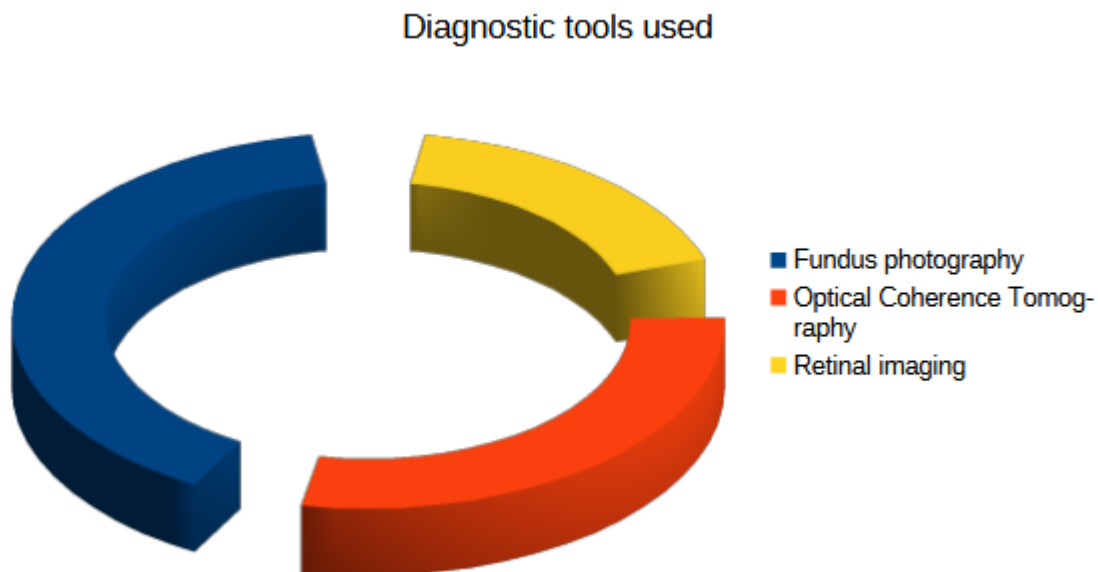


Figure 12 Diagnostic tools used

5. Discussion of results

5.1 General Public Data

5.1.1 Age Distribution and Screen Usage

From the age distribution (Figure 1), the majority of respondents fall within the 18–24 age group (75.6%), with 80.3% reporting that they do not wear glasses or contact lenses. However, 92.1% reported spending more than four hours daily on screens, which could contribute to eye strain and other vision issues.

Interpretation: Younger respondents are more likely to engage in prolonged screen use, possibly due to work, education, or social media habits. To address this, public awareness campaigns could focus on educating younger demographics about the risks of excessive screen time and the importance of regular breaks.

5.1.2 Eye Check-Up Habits

The data (Figure 5) shows that nearly half (47.2%) of the respondents never go for eye check-ups, and only 8.7% go once a year.

Interpretation: This indicates a lack of awareness or prioritization of regular eye health. Community outreach programs emphasizing the importance of preventative care could increase annual check-up rates.

5.1.3 Awareness of Pathological Myopia

From (figure 6) Around 51.2% of the general public had not heard of Pathological Myopia, while the remaining learned about it primarily through the internet (35.1%) and doctors (19.5%).

Interpretation: There is limited awareness of the condition, with digital platforms being the most effective communication tool. Health promotion efforts should utilize social media and online content to educate the public on eye conditions.

5.1.4 Eye Care Practices

From (figure 7) While 59.1% of respondents reported taking regular breaks from screens, only 13.4% visit eye doctors periodically, and 39.4% eat a nutrient-rich diet for eye health.

Interpretation: Although many practice screen breaks, holistic eye care practices are underutilized. Promoting balanced approaches, including dietary improvements and regular medical check-ups, is crucial.

5.2 Eye Specialist Data

5.2.1 Experience and Challenges

Most specialists (66.7%) had 1–3 years of experience, with 88.9% encountering Pathological Myopia in their practice (Figure 10). Challenges like a lack of equipment and patient delays were common themes in the qualitative response

Interpretation: The lack of diagnostic tools and late patient presentations hinder effective management. Investments in affordable diagnostic tools and public education campaigns to encourage early diagnosis are essential.

5.2.2 Symptoms and Risk Factors

Blurred vision (100%) and retinal detachment (66.7%) were frequently associated with Pathological Myopia, while prolonged near work (77.8%) and genetics (66.7%) were noted as significant risk factors.

Interpretation: These findings align with global studies emphasizing lifestyle and genetic predispositions as major contributors. Eye care professionals should emphasize lifestyle modifications to mitigate risks.

5.2.3 Effectiveness of Early Diagnosis

Specialists unanimously recognized early diagnosis as crucial, with 66.7% rating it as very effective.

Interpretation: This underscores the importance of timely interventions. Policies to integrate routine screenings into general health check-ups could enhance early detection rates.

6. Recommendations

In order to address the issues identified in the findings, we recommend the following actions:

6.1 Recommendations for the General Public

Increase Awareness about Pathological Myopia: In order to increase awareness in the general public we recommend launching public health campaigns on platforms such as social media, local television, and community gatherings to educate people about Pathological Myopia. These campaigns should target younger individuals, who are the most active screen users, and include content in local languages.

Promote Regular Eye Check-ups: In order to increase regular eye check-ups, we recommend organizing free or subsidized eye screening programs in schools, workplaces, and communities to encourage regular check-ups and early diagnosis.

Advocate for Healthy Screen Habits: We recommend developing mobile apps or tools that remind users to follow healthy screen habits, such as taking regular breaks, adhering to the 20-20-20 rule, and reducing blue light exposure on devices.

Encourage Healthy Lifestyles: We recommend emphasizing the importance of outdoor activities and eye-healthy diets through community outreach programs and school health initiatives.

6.2 Recommendations for Eye Care Professionals

Improve Access to Diagnostic Tools: In order to ease access to diagnostic tools by eye specialists we recommend collaborating with government bodies and NGOs to provide affordable and modern diagnostic tools, such as Optical Coherence Tomography (OCT) and fundus photography, especially in underserved areas.

Capacity Building and Training: In order to improve skill sets for Eye specialists we recommend funding continuous professional development programs for eye specialists to enhance their diagnostic and management skills for conditions like Pathological Myopia.

Enhance Patient Education: We recommend providing patients with accessible and simple materials that explain the importance of early diagnosis and treatment. Community seminars and workshops can complement these efforts.

6.3 Policy Recommendations

Integrate Eye Health into Primary Healthcare: We recommend mandating eye health as part of general health screenings in schools, workplaces, and clinics to enable early detection of eye conditions like Pathological Myopia.

Subsidize Treatment and Equipment Costs: We recommend introducing financial subsidies for diagnostic tools and treatment options to address economic barriers faced by both patients and specialists.

Incorporate Local Languages in Awareness Campaigns: We recommend translating awareness materials and campaigns into local languages to make them accessible to non-English-speaking communities.

7. Challenges Faced and Solutions

7.1 Challenges

Limited Awareness of Myopia: Lack of awareness about myopia and its long-term effects resulted in low participation rates.

Limited Access to Advanced Medical Equipment: We faced significant difficulties in accessing high-quality retinal imaging devices like OCT and fundus cameras to fully understand how pathological myopia is diagnosed.

Lack of Trained Ophthalmologists: There was a shortage of trained ophthalmologists to perform and annotate retinal examinations.

7.2 Solutions

Awareness campaigns: Launching awareness campaigns through local media, schools, and community centers to educate people about myopia and the importance of early detection.

Partnerships: Seeking partnerships with international medical organizations or NGOs and collaborating with universities and medical institutions to secure donations or funding for advanced equipment.

Training programs: Implementing training programs and workshops to build local expertise, collaborating with universities and medical institutions.

8. Conclusion

The integration of AI, particularly through advanced deep learning models, has demonstrated remarkable success in automating and enhancing the diagnosis of pathological myopia. While challenges persist,

particularly in low-resource settings, the ongoing advancements in datasets, algorithms, and implementation strategies indicate a promising trajectory. By addressing barriers to adoption, AI-driven solutions could revolutionize diagnostic practices, providing timely and accurate care for pathological myopia patients worldwide.

This research into pathological myopia practices in Kampala adopted a pragmatic philosophy, employing both quantitative and qualitative methods. It utilized a mixed-methods approach, incorporating quantitative data from surveys and qualitative data from interviews. This approach facilitated the examination of statistical trends as well as the exploration of the underlying context behind the data.

References

- [1] H. Fang *et al.*, “Open Fundus Photograph Dataset with Pathologic Myopia Recognition and Anatomical Structure Annotation,” *Sci Data*, vol. 11, no. 1, Dec. 2024, doi: 10.1038/s41597-024-02911-2.
- [2] A. Kalyanasundaram, “Detection of Pathological Myopia Using Convolutional Neural Network,” *International Journal of Psychosocial Rehabilitation*, vol. 24, no. 5, 2020, doi: 10.37200/ijpr/v24i5/pr201929.
- [3] Babarinde, O. (2020). AI adoption in healthcare: A comparative analysis between the USA and Africa. *Journal of Healthcare Technology*, 15(3), 45-57.
- [4] Chen, Y., Liu, X., & Wang, J. (2021). Optimizing model performance through transfer learning in medical imaging. *International Journal of Medical Informatics*, 148, 104387.
<https://doi.org/10.1016/j.ijmedinf.2021.104387>
- [5] Devda, R., & Eswari, J. (2019). Convolutional neural networks for pathological myopia image analysis. *Journal of Medical Imaging and Health Informatics*, 9(6), 1234-1240.
<https://doi.org/10.1166/jmihi.2019.2865>